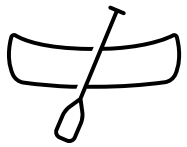
Group 1

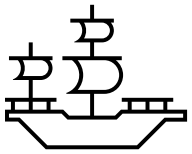
Lukai Wang, Tessel Ottens,
Victoria Vu, Chalmy Ceron
and Petrine Ladegaard



What learning theory is applied?



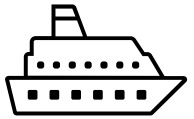
- Lens of researcher is cognitivist (DCog)



“The study uses a Distributed cognition (DCog) framework for analysis (Hutchins 1995).”

"...the studies outlines show how bodily actions and gestures have fundamental cognitive and communicative functions." (p.2318)

"While low-fidelity simulators are considered to be well suited for what is labelled "cognitive tasks", high-fidelity simulators are more appropriate when training motor skills according" (p.2315)



- Elements of socio-cultural/constructivism in the simulation

The students are assigned specific professional roles, such as the officer-of-the-watch and the lookout. The actual learning happens in the space between them, driven by their dialogue, teamwork, and collaborative problem-solving.



Research structure

- Theory (DCog) →open & exploratory question (hand gestures) →connection to theory →results

"This means that the boundaries between what is considered internal, mental processes and external, socio-material structures, are blurred and plastic; What happens inside a learner is the propagation of some kinds of structures of a system to another through mediation" (p. 2316)

The cases were not chosen to test a prediction/hypothesis, they were chosen to "demonstrate different aspects that emerged as interesting." (p.2319)

What did we learn from this exercise?



Keeping learning theories in mind when reading research papers



Being able to differentiate learning theories within one research paper



Comprehensive reading with preparation